

EX:NO:19**PROGRAM:**

```
def bubble_sort(arr):
```

```
    n = len(arr)
```

```
    for i in range(n):
```

```
        swapped = False
```

```
        for j in range(0, n-i-1):
```

```
            if arr[j] > arr[j+1]:
```

```
                arr[j], arr[j+1] = arr[j+1], arr[j]
```

```
                swapped = True
```

```
        if not swapped:
```

```
            break
```

```
    return arr
```

```
# Tests
```

```
print(bubble_sort([64,25,12,22,11])) # [11,12,22,25,64]
```

```
print(bubble_sort([29,10,14,37,13])) # [10,13,14,29,37]
```

```
print(bubble_sort([3,5,2,1,4]))      # [1,2,3,4,5]
```

```
print(bubble_sort([1,2,3,4,5]))      # [1,2,3,4,5]
```

```
print(bubble_sort([5,4,3,2,1]))      # [1,2,3,4,5]
```

Output

```
[11, 12, 22, 25, 64]
```

```
[10, 13, 14, 29, 37]
```

```
[1, 2, 3, 4, 5]
```

```
[1, 2, 3, 4, 5]
```

```
[1, 2, 3, 4, 5]
```

```
=== Code Execution Successful ===
```